UNIT 1 – MINI PROJECT REPORT

Enterprise Student Record and Academic Management Platform

Design Thinking and Product Planning

Course: DevOps & Cloud Foundation (IBM)

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Abstract

Higher-education institutions continue to rely, in large part, on fragmented and often manual processes to manage core academic activities such as admissions, course registration, attendance capture, examinations, and result declaration. These fragmented workflows lead to data duplication, delayed reporting, error-prone record keeping, and a poor experience for students, faculty, and administrative staff alike. This report presents the Unit 1 deliverable for the DevOps & Cloud Foundation (IBM) course: the business understanding, design-thinking exploration, and product-planning artifacts for a proposed Enterprise Student Record and Academic Management Platform.

The work follows the Design Thinking methodology demonstrated by the trainer, applying its five stages — Empathize, Define, Ideate, Prototype, and Test — to the academic-management domain, and layers on IBM's Enterprise Design Thinking principles of Focus on User Outcomes, Restless Reinvention, and Diverse Empowered Teams, expressed through the Observe → Reflect → Make loop. Primary and secondary stakeholders were identified and analyzed, and representative user personas were developed for students, faculty members, academic administrators, examination-cell staff, and IT support personnel to ground requirements in real user goals and pain points.

From this foundation, the report derives a Business Requirements Document, a Functional Requirements Specification, and a Non-Functional Requirements Specification, followed by a Product Vision Document articulating the platform's vision, mission, success criteria, and business objectives. A Minimum Viable Product is scoped to deliver the highest-value capabilities first, and a phased Product Roadmap sequences subsequent releases in alignment with the broader course progression. An initial high-level system architecture is proposed, together with a Project Charter, a Git-based repository initialization plan, a recommended project folder structure, and a summary of product-discovery activities and findings.

Collectively, these artifacts constitute a complete, agile-ready product plan: a well-understood problem space, a validated set of user needs, a scoped and prioritized product backlog direction, and a technical and organizational foundation from which subsequent units — covering Agile planning, architecture, implementation, and DevOps practices — can proceed with minimal ambiguity and maximum stakeholder alignment.

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1. Objective

The objective of this Unit 1 mini project is for students to independently apply the Design Thinking methodology, as demonstrated by the trainer, to design a comprehensive software solution for managing student records and academic information within a higher-education institution. Rather than beginning with coding, the focus of this unit is on understanding the educational domain, identifying stakeholders, analyzing user needs, defining software requirements, and preparing a product vision and roadmap. By the end of this unit, the outcome is a well-planned software product — the Enterprise Student Record and Academic Management Platform — that is ready to move into Agile planning and development in subsequent units.

- Understand the academic management domain and its current, largely manual, pain points.
- Apply human-centered Design Thinking and IBM Enterprise Design Thinking to reframe the problem around real user outcomes.
- Identify and analyze stakeholders, and build user personas that ground product decisions in real needs.
- Translate insight into concrete, testable functional and non-functional requirements.
- Define a product vision, an MVP, and a phased roadmap, and prepare the project for Agile execution.

2. Business Understanding

This section documents the study of the existing academic management process across a typical higher-education institution, the key sub-processes involved, and the problems associated with manual or semi-digital record management that motivate the proposed platform.

2.1 Overview of the Academic Management Process

Academic management spans the complete lifecycle of a student's association with an institution — from admission through graduation — and the parallel administrative, teaching, and examination workflows that support it. In most institutions today, this lifecycle is handled through a combination of spreadsheets, paper forms, standalone legacy tools, and email, each owned by a different department (admissions office, academic office, examination cell, individual faculty members). The proposed platform consolidates these fragmented touchpoints into a single, role-based system of record.

2.2 Core Sub-Processes Studied

2.2.1 Student Admissions

Covers enquiry handling, application submission, eligibility verification, document collection, fee payment confirmation, seat allotment, and enrollment confirmation. Today this is largely paper- and email-driven, with eligibility checks performed manually against scanned mark-sheets.

2.2.2 Course Registration

Covers semester-wise subject selection (core and elective), section/batch allocation, prerequisite validation, and registration confirmation. Manual registration frequently results in section overloads, missed prerequisite checks, and duplicate entries across departmental spreadsheets.

2.2.3 Attendance Management

Covers day-wise and subject-wise attendance capture by faculty, aggregation into monthly and semester-wise percentages, and eligibility flags for examinations based on minimum attendance thresholds. Manual attendance registers are slow to consolidate and prone to transcription errors.

2.2.4 Examination Management

Covers exam scheduling, seating arrangement, hall-ticket generation, invigilation duty allocation, and answer-script tracking. Coordinating this across departments via spreadsheets and circulars is time-consuming and error-prone, particularly during clashes in exam scheduling.

2.2.5 Result Management

Covers marks entry by faculty/examination cell, moderation, grade computation, transcript and mark-sheet generation, and result publication. Manual aggregation of marks across multiple subjects and evaluators is a major source of delay and disputed results.

2.3 Problems Associated with Manual Record Management

Problem Area	Description	Impact
Data Duplication	Student data re-entered separately by admissions, academics, and examination departments.	Inconsistent records, wasted effort
Delayed Reporting	Attendance and result aggregation performed manually at period end.	Late interventions for at-risk students
Human Error	Manual transcription of marks and attendance across registers/sheets.	Disputed results, rework, grievances
Limited Accessibility	Records siloed in individual departments/desktops.	Poor visibility for students and parents
No Audit Trail	Paper records offer no change history or accountability.	Compliance and audit risk
Scalability Limits	Manual processes do not scale with rising enrollment.	Administrative overload each semester

Table 2.1 — Problems associated with manual academic record management

3. Design Thinking Report

Design Thinking was applied as the core problem-solving methodology for this project, following its five canonical stages. Each stage was used to progressively narrow an initially broad, ambiguous problem space — "academic records are hard to manage" — into a specific, buildable, and testable product concept.

3.1 Empathize

The empathize stage focused on understanding the lived experience of everyone who touches the academic-record lifecycle. Informal interviews and process walkthroughs (simulated for this exercise using domain research and institutional documentation) were used to surface frustrations and unmet needs.

- Students struggle to track attendance percentage and results in real time and rely on notice boards or word of mouth.
- Faculty spend disproportionate time on clerical work — attendance registers, marks entry, report compilation — rather than teaching.
- Administrators lack a single, real-time view of enrollment, attendance, and academic performance across departments.
- Examination cell staff manage scheduling and result publication through spreadsheets shared over email, risking version conflicts.

3.2 Define

Insights from the empathize stage were synthesized into a clear problem statement:

> *"Students, faculty, and administrative staff at higher-education institutions need a unified, role-based platform to manage admissions, registration, attendance, examinations, and results — because today's fragmented, manual processes cause data duplication, delayed reporting, and avoidable errors that affect academic outcomes and institutional efficiency."*

3.3 Ideate

A wide range of solution ideas were brainstormed and then filtered for feasibility, desirability, and business viability. Selected ideas that shaped the product concept include:

- A single web-based platform with distinct, role-based dashboards for students, faculty, and administrators.
- Digital attendance capture with automatic percentage computation and low-attendance alerts.
- Centralized marks entry with built-in moderation workflow and automatic grade/transcript generation.
- Self-service student portal for registration, attendance visibility, and result access.
- Notification engine (email/SMS/in-app) for deadlines, low attendance, and result publication.
- Analytics and reporting dashboard for administrators covering enrollment, attendance trends, and academic performance.

3.4 Prototype

Low-fidelity prototyping for Unit 1 takes the form of structural and conceptual artifacts rather than pixel-perfect screens: the initial system architecture diagram (Section 12), the MVP definition (Section 10), and the functional requirements (Section 7) together act as a conceptual prototype of the platform, describing what

the first testable version will do and how its major components will interact, ahead of visual/UI prototyping in later units.

3.5 Test

Because Unit 1 precedes implementation, the "test" stage is scoped as a validation-planning activity: the problem statement, personas, and MVP scope defined in this report are earmarked for review with representative stakeholders (a sample of students, a faculty member, and an administrative staff member) before Agile development begins in later units, so that assumptions are validated before engineering effort is committed.

4. IBM Design Thinking Report

In addition to the classical five-stage Design Thinking process, this project applies IBM's Enterprise Design Thinking framework, which is built around three core principles and a continuous practice loop suited to teams working at enterprise scale and speed.

4.1 A Focus on User Outcomes

Every feature considered for the platform is tied back to a measurable user outcome rather than a technical capability for its own sake. For example, "digital attendance capture" is framed not as a feature but as enabling the outcome "students and advisors can identify and act on attendance risk before it becomes an examination-eligibility problem."

4.2 Restless Reinvention

The product plan is treated as a living document. The MVP scope, roadmap, and requirements captured in this report are the current best understanding, explicitly expected to be revisited as user feedback and Agile sprint learnings accumulate in later units, rather than being treated as a final, unchangeable specification.

4.3 Diverse Empowered Teams

Although this mini project is completed individually for Unit 1, the product plan is written to support a small, cross-functional team model — combining product, design, and engineering perspectives — that would take ownership of the platform end-to-end during subsequent Agile units, with each role empowered to make decisions within their domain rather than escalating every decision upward.

4.4 The Loop — Observe, Reflect, Make

IBM's Loop is the operating rhythm through which the three principles above are put into practice on an ongoing basis. The table below documents how the Loop was applied within this Unit 1 exercise.

Loop Stage	Activity in this Project
Observe	Studied the academic management process end-to-end (Section 2); identified stakeholders and their goals, responsibilities, and challenges (Sections 5–6).
Reflect	Synthesized observations into a single problem statement (Section 3.2), a prioritized set of pain points, and success criteria for the product (Section 9).
Make	Translated reflection into concrete artifacts: BRD, FRS, NFRS, product vision, MVP, roadmap, and initial architecture (Sections 7–13) — the tangible outputs the team can act on next.

Table 4.1 — Observe → Reflect → Make cycle applied to this project

This Loop is intended to repeat continuously through subsequent units: each sprint's outcomes will be observed, reflected upon, and used to re-make the plan, rather than executing this Unit 1 plan unchanged through to project completion.

5. Stakeholder Analysis Report

Stakeholders were identified across primary users (who interact with the platform directly and regularly), secondary users (who interact with it occasionally or through others), and external/support stakeholders (who influence or are influenced by the system without using it directly).

Stakeholder	Category	Interest / Stake	Influence
Students	Primary	Registration, attendance visibility, timely and accurate results	Medium
Faculty Members	Primary	Fast attendance/marks entry, less clerical overhead	High
Academic Administrators	Primary	Institution-wide visibility, accurate reporting, compliance	High
Examination Cell Staff	Primary	Reliable scheduling, moderation workflow, timely publication	High
IT Support Personnel	Secondary	System uptime, ease of maintenance, secure access management	Medium
Parents / Guardians	Secondary	Visibility into attendance and results	Low
Institution Leadership	External	Institutional reputation, regulatory compliance, cost efficiency	High
Regulatory Bodies	External	Accreditation, data retention and reporting standards	Medium

Table 5.1 — Stakeholder analysis matrix

Faculty members, academic administrators, and examination cell staff are treated as high-influence, high-interest stakeholders and are managed as close, continuously-consulted partners through the product's development. Students, while high in number, are treated as high-interest but medium-influence in institutional decision-making, and are engaged primarily through usability testing and feedback rather than governance.

6. User Persona Report

Three detailed personas were developed, representing the platform's primary user groups, to keep requirements and design decisions anchored in concrete goals, responsibilities, challenges, and expectations rather than abstract user categories.

Persona 1: Riya Sharma — Second-Year Undergraduate Student

"I just want to know my attendance and results without chasing three different offices."

Goals	Register for courses smoothly each semester; stay above the attendance threshold; access results as soon as they are published.
Responsibilities	Attend classes, submit assignments, monitor own attendance and academic progress.
Challenges	No real-time visibility into attendance percentage; result delays and unclear grievance process; registration conflicts with prerequisites.
Expectations from the Platform	A simple self-service portal showing live attendance, registered courses, and results, with alerts before attendance drops below the eligibility threshold.

Persona 2: Prof. Anand Deshmukh — Faculty Member, Department of Computer Science

"Marking attendance and entering marks eats into the time I'd rather spend on teaching and research."

Goals	Deliver quality instruction; complete administrative record-keeping quickly and accurately; identify at-risk students early.
Responsibilities	Take daily/subject-wise attendance, enter internal and examination marks, generate progress reports for advisees.
Challenges	Duplicate data entry across register, spreadsheet, and office submissions; no easy way to flag struggling students to administration.
Expectations from the Platform	A fast, mobile-friendly attendance and marks-entry workflow, automatic aggregation, and a dashboard highlighting at-risk students in advised sections.

Persona 3: Meera Kulkarni — Academic Administrator / Examination Cell Coordinator

"I need one dashboard that tells me what's happening across every department, not five spreadsheets that never agree."

Goals	Maintain accurate institution-wide academic records; ensure exam scheduling and result publication run without conflicts; produce compliance reports quickly.
Responsibilities	Oversee admissions data consolidation, coordinate exam schedules and invigilation, moderate and publish results, generate institutional reports.
Challenges	Reconciling inconsistent departmental spreadsheets; manual report compilation for leadership and regulators; limited audit trail for corrections.
Expectations from the Platform	A single source of truth for enrollment, attendance, and results with role-based access, an audit trail, and one-click institutional reporting.

7. Business Requirements Document (BRD)

7.1 Purpose

This Business Requirements Document defines the business-level needs that the Enterprise Student Record and Academic Management Platform must satisfy. It serves as the bridge between the business understanding and stakeholder analysis captured earlier in this report and the detailed functional and non-functional specifications that follow.

7.2 Business Objectives

- Consolidate admissions, registration, attendance, examination, and result data into a single system of record.
- Reduce administrative turnaround time for reporting and result publication.
- Improve data accuracy and reduce disputes arising from manual transcription errors.
- Provide real-time visibility to students, faculty, and administrators appropriate to their role.
- Establish a scalable technical and process foundation for future institutional growth.

7.3 Business Scope

In scope for the platform: student admissions record-keeping, course registration, attendance capture and reporting, examination scheduling support, marks entry and moderation, result generation and publication, and role-based reporting for administrators.

Out of scope for the initial release: online fee payment processing, learning content delivery (LMS functionality), hostel/transport management, and alumni relationship management. These are noted as candidates for future roadmap phases (Section 11).

7.4 Business Rules

ID	Business Rule
BR-01	A student must maintain a minimum of 75% attendance per subject to be eligible for the end-semester examination, unless a documented exemption is approved by the administrator.
BR-02	Course registration for a semester must be validated against prerequisite completion before confirmation.
BR-03	Marks entered by faculty require a moderation/approval step by the examination cell before results are published.
BR-04	Every change to an academic record must be logged with the user, timestamp, and nature of the change.
BR-05	Only users with the appropriate role may access, enter, or modify a given category of record (role-based access control).

Table 7.1 — Core business rules

8. Functional Requirements Specification (FRS)

Functional requirements describe what the system must do. They are grouped by module and each is assigned a unique identifier and a priority (Must / Should / Could, using MoSCoW prioritization) for use in later Agile planning units.

8.1 Admissions & Enrollment

ID	Requirement	Priority
FR-01	The system shall allow administrators to create and manage student admission records, including personal, academic, and document details.	Must
FR-02	The system shall verify applicant eligibility against configurable admission criteria.	Must
FR-03	The system shall generate a unique enrollment/student ID upon confirmed admission.	Must

8.2 Course Registration

ID	Requirement	Priority
FR-04	The system shall allow students to register for core and elective subjects each semester.	Must
FR-05	The system shall validate prerequisite completion before confirming registration.	Must
FR-06	The system shall prevent registration beyond a configurable section/batch capacity.	Should

8.3 Attendance Management

ID	Requirement	Priority
FR-07	The system shall allow faculty to record subject-wise attendance per session.	Must
FR-08	The system shall automatically compute running attendance percentage per student per subject.	Must
FR-09	The system shall notify students and advisors when attendance falls below the configured eligibility threshold.	Should

8.4 Examination Management

ID	Requirement	Priority
FR-10	The system shall allow the examination cell to create and publish exam schedules.	Must
FR-11	The system shall generate hall tickets for eligible, registered students.	Should
FR-12	The system shall support invigilation duty allocation to faculty.	Could

8.5 Result Management

ID	Requirement	Priority
FR-13	The system shall allow faculty to enter internal and examination marks per subject.	Must
FR-14	The system shall support a moderation/approval workflow before result publication.	Must
FR-15	The system shall automatically compute grades and generate mark-sheets/transcripts upon publication.	Must
FR-16	The system shall notify students automatically when results are published.	Should

8.6 Reporting & Administration

ID	Requirement	Priority
FR-17	The system shall provide administrators with dashboards summarizing enrollment, attendance, and result trends.	Should
FR-18	The system shall allow export of institutional reports in common formats (PDF/CSV).	Should
FR-19	The system shall maintain a full audit trail of record changes for compliance purposes.	Must

Table 8.1–8.6 — Functional requirements grouped by module, prioritized using MoSCoW

9. Non-Functional Requirements Specification (NFRS)

Non-functional requirements describe the quality attributes and constraints the platform must satisfy, independent of specific features.

Category	Requirement	Target / Metric
Security	All access shall be authenticated and authorized using role-based access control; sensitive data shall be encrypted in transit and at rest.	TLS 1.2+, AES-256
Usability	The interface shall be usable by non-technical staff and students with minimal training, and shall be responsive across desktop and mobile browsers.	SUS score ≥ 75
Reliability	The platform shall remain available during critical academic periods (registration, exams, results) with graceful degradation on partial failure.	99.5% uptime
Scalability	The platform shall support growth in student enrollment and concurrent usage during peak periods (e.g., result day) without redesign.	10,000+ concurrent users
Performance	Core pages (dashboard, attendance entry, results) shall load within an acceptable response time under normal load.	< 2s page load
Maintainability	The system shall be built with modular services and documented APIs to support ongoing enhancement by a small team.	Modular / documented
Auditability	All create/update/delete actions on academic records shall be logged with actor, timestamp, and change details.	100% coverage
Compliance	Data retention and privacy handling shall align with institutional and regulatory record-keeping requirements.	Policy-aligned

Table 9.1 — Non-functional requirements

10. Product Vision Document

10.1 Vision Statement

> *"For students, faculty, and administrators at higher-education institutions who are burdened by fragmented, manual academic record-keeping, the Enterprise Student Record and Academic Management Platform is a unified, role-based system that consolidates admissions, registration, attendance, examinations, and results into a single, reliable source of truth — unlike disconnected spreadsheets and paper registers, our platform gives every stakeholder real-time, trustworthy visibility into the information that matters to them."*

10.2 Mission Statement

To simplify and digitize academic administration for higher-education institutions, reducing manual effort and error while giving students, faculty, and administrators timely, accurate, and role-appropriate access to academic information.

10.3 Success Criteria

- Reduction in time taken to compile and publish semester results after examinations.
- Reduction in attendance- and marks-related discrepancy complaints.
- Increase in the proportion of students who can self-serve attendance/result queries without contacting an office.
- Positive usability feedback from a representative sample of students, faculty, and administrators.
- Successful, uneventful handling of peak-load events such as registration week and result publication day.

10.4 Business Objectives

- Deliver a working MVP covering the core student-record lifecycle within the current academic term.
- Establish a maintainable technical foundation that supports Agile, sprint-based delivery in subsequent units.
- Position the institution to retire the most error-prone manual workflows first (attendance and results).

11. Minimum Viable Product (MVP) Definition

The MVP is scoped to deliver the smallest set of capabilities that lets the institution stop relying on manual processes for the highest-pain workflows — attendance and results — while establishing the record-keeping backbone (student and course records) that every later feature depends on.

11.1 MVP Scope

Module	Included in MVP
Student Registration	Create/maintain student profile and enrollment record; assign unique student ID.
Course Registration	Semester-wise subject registration with prerequisite validation.
Attendance	Subject-wise attendance capture by faculty; automatic percentage computation; low-attendance flag.
Academic Report Generation	Consolidated student report (attendance + marks) viewable by student, faculty, and admin per role.

Table 11.1 — First release (MVP) scope

11.2 Explicitly Deferred from MVP

- Full examination scheduling and hall-ticket generation.
- Marks moderation workflow and formal transcript generation.
- Notification engine (email/SMS).
- Administrator analytics dashboards and export tooling.

These are sequenced into later roadmap phases below, once the MVP validates the core record-keeping and attendance workflow with real users.

12. Product Roadmap

The roadmap is phased to align with the progression of the DevOps & Cloud Foundation course, so that each phase's scope maps naturally onto the Agile planning, architecture, and delivery units that follow this one.

Phase	Theme	Key Deliverables
Phase 1 (MVP)	Core Records & Attendance	Student & course registration, attendance capture and reporting, basic role-based dashboards.
Phase 2	Examinations & Results	Exam scheduling, marks entry, moderation workflow, grade computation, result publication.
Phase 3	Engagement & Insight	Notification engine, administrator analytics dashboards, exportable institutional reports.
Phase 4	Scale & Integration	Performance hardening for peak load, integration with payment/LMS/biometric systems, expanded audit and compliance tooling.

Table 12.1 — Phased product roadmap

Each phase is intended to correspond to one or more Agile sprints once the project moves into subsequent DevOps & Cloud Foundation units, with Phase 1 forming the initial sprint backlog.

13. Initial System Architecture Diagram

The diagram below presents the proposed high-level, layered architecture for the platform's first release. It separates concerns into a presentation layer, an API/access layer enforcing authentication and authorization, an application/business-logic layer organized by module, an integration layer for external systems, and a data layer for persistence — a structure chosen for clarity and maintainability by a small team, and one that maps cleanly onto containerized, cloud-deployable services in later DevOps units.

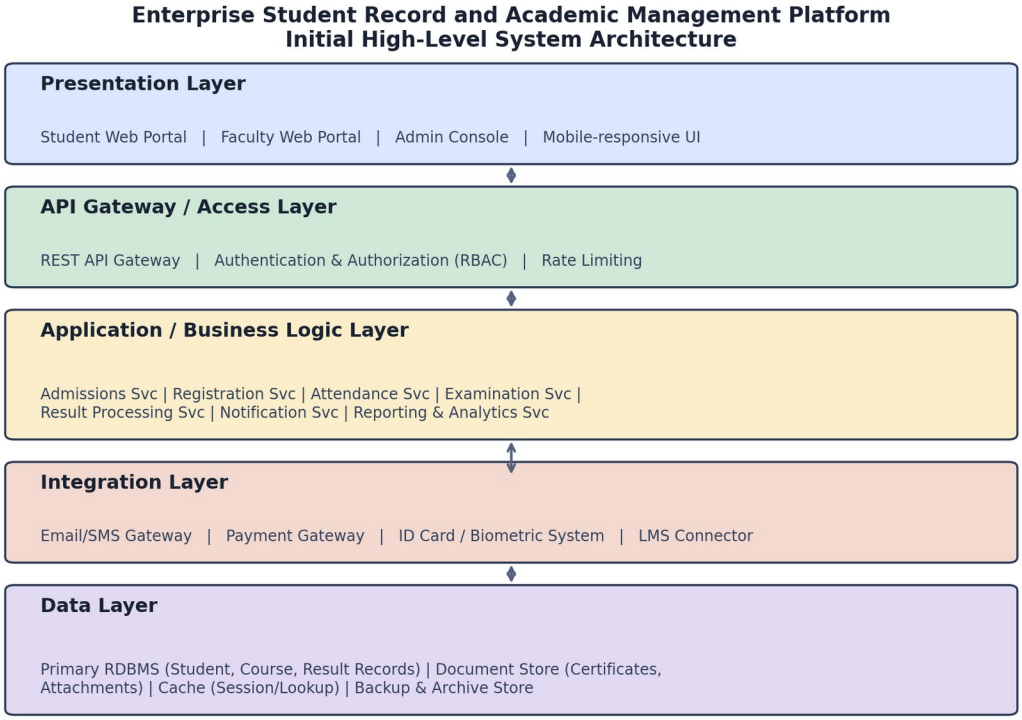


Figure 13.1 — Initial high-level system architecture

13.1 Layer Descriptions

- Presentation Layer: role-based web portals for students, faculty, and administrators, built responsively for desktop and mobile access.
- API Gateway / Access Layer: single entry point enforcing authentication, role-based authorization, and rate limiting ahead of all business services.
- Application / Business Logic Layer: modular services aligned to the FRS modules — admissions, registration, attendance, examination, results, notifications, and reporting.
- Integration Layer: connectors to external systems such as email/SMS gateways, payment gateways, ID/biometric systems, and a future LMS.
- Data Layer: a primary relational store for structured academic records, a document store for certificates/attachments, a cache for session and lookup data, and a backup/archive store.

14. Project Charter

Project Name	Enterprise Student Record and Academic Management Platform
Project Sponsor	Institution Academic Office (represented for this exercise by the course trainer)
Project Owner / Author	Soham Sable (Department of AIDS)
Business Case	Replace fragmented, manual academic record-keeping with a unified digital platform to reduce error, delay, and administrative overhead (see Section 2 and 7).
Objectives	Deliver a phased platform (Section 12) starting with an MVP (Section 11) covering student records and attendance.
Scope	As defined in the BRD (Section 7.3): admissions, registration, attendance, examinations, results, and reporting; excludes payments, LMS, hostel/transport for the initial release.
Key Stakeholders	Students, faculty, academic administrators, examination cell staff, IT support (Section 5).
Constraints	Delivery aligned to course unit timeline; small-team build; must integrate with existing institutional identity and communication channels in later phases.
Assumptions	Stakeholders are available for validation; institution will provide access to representative process documentation; cloud infrastructure will be available for later deployment units.
High-Level Milestones	Unit 1: Business understanding & product plan (this report). Unit 2+: Agile backlog, architecture detail, implementation, DevOps pipeline.
Success Measures	As defined in Section 10.3 (Success Criteria).

Table 14.1 — Project charter summary

15. Project Repository Initialization

The project repository was initialized using Git to establish version control from the outset of the project, ahead of any code being written, in line with DevOps practice of treating infrastructure and history as first-class artifacts from day one.

15.1 Initialization Steps

- git init — initialize a local repository for the project.
- Create the baseline folder structure (Section 16) and an initial README.md describing the project.
- git add . && git commit -m "Initial commit: Unit 1 business understanding and product planning" — first commit capturing the planning artifacts.
- Create a remote repository named "devops & cloud Unit1 mini project" and link it with git remote add origin <repository-url>.
- git push -u origin main — publish the initial commit to the remote repository.

Subsequent units will build on this repository using feature branches per module (e.g., feature/attendance, feature/registration) merged via pull requests, in preparation for the Agile and DevOps practices introduced later in the course.

16. Project Folder Structure

The following top-level folder structure is proposed for the repository, separating planning documentation, source code (to be added in later units), and supporting assets:

```
devops-cloud-unit1-mini-project/
├── docs/
│   ├── unit1-mini-project-report.pdf        (this report)
│   ├── brd.md / frs.md / nfrs.md
│   ├── personas/
│   └── architecture/
│       └── initial-architecture-diagram.png
├── src/                                      (reserved for later units)
│   ├── frontend/
│   ├── backend/
│   └── services/
├── tests/                                    (reserved for later units)
├── infra/                                    (reserved for DevOps/CI-CD units)
├── .gitignore
├── README.md
└── LICENSE
```

The docs/ folder holds this Unit 1 report and its constituent artifacts; src/, tests/, and infra/ are created now as placeholders so the repository structure is ready to receive code, automated tests, and CI/CD configuration without restructuring in later units.

17. Product Discovery Documentation

This section summarizes the discovery activities carried out to arrive at the product plan documented in this report, and the key findings that most influenced scope and prioritization decisions.

17.1 Discovery Activities

- Domain research into typical academic-management workflows across admissions, registration, attendance, examinations, and results.
- Stakeholder identification and analysis (Section 5), informing whose needs would drive prioritization.
- Persona development (Section 6) to translate stakeholder categories into concrete, relatable user profiles.
- Design Thinking workshops-in-miniature (Section 3), moving from empathy through to a testable problem statement and solution ideas.
- Requirements elicitation, producing the BRD, FRS, and NFRS (Sections 7–9).

17.2 Key Discovery Findings

- Attendance and result management are the two highest-friction, highest-frequency workflows for both students and faculty — this directly shaped the MVP scope (Section 11).
- Administrators' primary unmet need is a single reliable source of truth rather than any one specific feature — reinforcing the product vision (Section 10.1).
- Role-based access and an audit trail were repeatedly implicated in trust and compliance concerns, elevating them from "nice to have" to explicit non-functional requirements (Section 9).
- A phased rollout starting with record-keeping and attendance, before examinations/results, reduces risk by validating the platform on lower-stakes workflows first.

18. Conclusion

This report has documented the complete Unit 1 deliverable for the Enterprise Student Record and Academic Management Platform: a grounded understanding of the academic-management business domain, a Design Thinking and IBM Enterprise Design Thinking exploration of the problem space, detailed stakeholder and persona analysis, a full set of business, functional, and non-functional requirements, a clear product vision, a scoped MVP and phased roadmap, an initial system architecture, and the project-management and repository foundations needed to begin Agile delivery.

Together, these artifacts show that the proposed platform is not simply a digitization of existing manual processes, but a deliberate redesign around the real outcomes that students, faculty, and administrators need — timely, trustworthy, role-appropriate access to academic information. The project is now well-positioned to proceed into Agile sprint planning, detailed architecture, and implementation in the subsequent units of the DevOps & Cloud Foundation course.

18.1 Deliverables Checklist

- Business Requirements Document (BRD) — Section 7
- Product Vision Document — Section 10
- Stakeholder Analysis Report — Section 5
- User Persona Report — Section 6
- Functional Requirements Specification (FRS) — Section 8
- Non-Functional Requirements Specification (NFRS) — Section 9
- Design Thinking Report — Section 3
- IBM Design Thinking Report — Section 4
- IBM Loop Documentation — Section 4.4
- MVP Definition — Section 11
- Product Roadmap — Section 12
- Initial System Architecture Diagram — Section 13
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